# Ziyang Zhang

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*Ph.D. student at Tsinghua EE. Research: time-series LLMs & world models, speech generation (TTS, speech LLMs), LLM alignment (DPO, weak-to-strong).*

## Education

**Tsinghua University** — Beijing, China
*Ph.D. in Electronic Engineering (Advisor: Assoc. Prof. Chao Zhang)* — *Aug. 2024 – Present*

**Tsinghua University** — Beijing, China
*B.Eng. in Electronic Engineering; Outstanding Graduate (2024); Academic Excellence Scholarship ×3* — *Sept. 2020 – July 2024*

## Research Internships

**Shanghai Artificial Intelligence Laboratory** — Nov. 2024 – Oct. 2026
*Research Intern, time-series team of the Intern-S model series. All time-series work below was done here.*

**Intern-S1: A Scientific Multimodal Foundation Model** (241B MoE) *Lead of the time-series modality* — Tech. report 2025

*Motivation* Much scientific data comes as raw signals (seismic and gravitational waves, light curves, EEG), sampled from once a day to GHz rates. The goal was a foundation model that reads them directly, as a modality alongside images.

*Approach* Added a time-series modality to a 28B-activated / 241B-total MoE model continually pre-trained on 5T tokens: adaptive downsampling and Transformer blocks compress raw signals of any sampling rate into representations the LLM can reason over, trained jointly with text and vision.

*My role* Responsible for the overall architecture and training of the time-series modality: encoder design, cross-modal alignment between the encoder and the LLM, and SFT. Ran the integration training of the time-series modality into the 241B model on the lab's XTuner framework.

**Intern-S2-Preview: A Scientific Agentic Foundation Model** (397B) *Time-series lead; sole owner of forecasting* — Tech. report 2026

*Motivation* Intern-S2 had to handle longer, higher-frequency signals and add forecasting: read an instruction in dialogue, choose the horizon and output future values.

*Approach* A long-sequence encoder with Q-Former compressive patching and a channel-wise Transformer raised the maximum input over Intern-S1-Pro from about 240k to 300k time steps, with 5–6× faster inference and about 20% of the GPU memory at maximum length. A new forecasting branch uses a Q-Former to gather LLM context and encoder features, conditions a causal Transformer forecaster through cross-attention, and predicts the horizon from the instruction. On SciTS it beats Intern-S1-Pro on 7 of 9 shared understanding tasks (PHU01 F1 36.8 → 66.9) and outperforms TimeOmni and specialized forecasters on most forecasting tasks, with 99% horizon accuracy.

*My role* Responsible for the overall architecture and training (integration) of the time-series modality; solely responsible for the architecture exploration of the new forecasting capability, from scratch.

**MMTA & TimeOmni-v: Multimodal Temporal Analysis with Time Series, Text, and Vision** *First author* — **NeurIPS 2026**

*Motivation* Benchmarks pair time series with text or with plots of the same signal, so whether MLLMs can combine them with real visual evidence was untested; judging whether a robot finished a task needs the instruction, video and joint traces together.

*Approach* Built MMTA, a tri-modal benchmark across nine domains under video+TS and image+TS layouts, with a 3,524-sample test split for classification and forecasting and success-weighted metrics. Eight MLLMs share one bottleneck: time series written as text give very long token sequences and weak alignment with vision. TimeOmni-v adds a time-series encoder and a forecasting head to Qwen2.5-Omni so that it handles time series natively, with a time-series RoPE that places series and video on one timeline and series tokens interleaved with video frames. It reaches an average swF1 of 0.601 vs. 0.442 for the best baseline, Gemini-3-Flash, and on a long-video robotics task cuts time-series tokens 232.8× and latency 23×.

**SciTS & TimeOmni: Scientific Time Series Understanding and Generation** *Second author* — **ICLR 2026**

*Motivation* Scientific series reach ten million points and 10 MHz; as text they are too long, as images they lose precision. The goal was a benchmark for understanding and generation, and a model that treats time series as a modality of its own.

*Approach* SciTS covers 12 disciplines, 43 understanding and generation tasks, and 54,023 instances. Across 17 models, general LLMs generalize better than specialized time-series models. TimeOmni, a unified time-series LLM for understanding and generation, ranks first on average in both.

*My role* Solely responsible for TimeOmni's architecture and training. A router sends each signal by length to a different time-series encoder, so one model handles series of very different lengths; encoded patches are reprogrammed into the LLM's representation space, and forecasting heads output values for generation tasks. Trained understanding and generation jointly on Qwen3-8B.

**TimeOmni-o1: Bidirectionally Coupling TSFMs and LLMs for Multimodal Reasoning and Forecasting** *First author* — Preprint

*Motivation* Series often come with text, such as an event report, and both understanding and forecasting need reasoning over the two. LLMs read series as digit tokens and handle long ones poorly; time-series foundation models (TSFMs) forecast well but cannot read or write text; time-series LLMs forecast on the LLM side and leave the TSFM's forecasting ability unused.

*Approach* TimeOmni-o1 couples a pretrained TSFM and an LLM in both directions: the TSFM passes a compressed encoding of the series to the LLM, which reasons in text; the reasoning is fed back into the TSFM, which forecasts conditioned on it. Built MMTR, a reasoning-annotated dataset with 500k+ samples across 12 domains. Understanding averages 84.2 (best baseline 74.9) and forecasting CRPS is 0.116, the lowest in every domain and better than the inner LLM or TSFM used alone; it also beats the dedicated time-series models on three open benchmarks.

**Time-JEPA: Grounding Video World Models with Physical Time Series** *Co-first author*

*Motivation* World models are trained mainly on video, but the real world also contains physical quantities over time, such as speeds, joint angles and forces, and the physical laws behind them. Physical laws such as Newton's are built on precise measurement, so a world model needs precise, dynamic representations of these quantities to understand them; models trained on video alone capture semantics well but miss such representations. The goal was to bring physical time series into world-model training.

*Approach* Time-JEPA fine-tunes V-JEPA with temporally aligned, low-dimensional physical time series as auxiliary supervision; evaluation uses video only. Across driving, robot manipulation and human motion, the model fits physical laws better (kinematic reconstruction trajectory error 4.150 m $\rightarrow$ 1.858 m), improves current- and future-state prediction under unseen conditions and unseen state variables, is more sensitive to physical anomalies, and generalizes to unseen scenarios.

*My role* Proposed the idea; designed the overall and core methods, chose the datasets and set the expected conclusions; supervised junior students who ran the experiments.

**Apple, Siri China R&D** Jan. 2024 – Oct. 2024

*Research Intern, speech generation*

**BELLE: Bayesian Speech Synthesizers Can Learn from Multiple Teachers** *First author* arXiv 2025

*Motivation* Text-to-speech is a one-to-many mapping with intrinsic uncertainty, but continuous-valued autoregressive TTS relies on a fixed-variance prior and yields a static point estimate that ignores the variability of natural speech. The goal was to model this uncertainty explicitly with Bayesian methods, without extra parameters or inference latency.

*Approach* BELLE brings Bayesian evidential learning into continuous-valued autoregressive TTS: a Normal-Inverse-Gamma distribution per acoustic frame, with variance predicted from context, captures data-dependent aleatoric uncertainty. Training data combine real recordings (about 706 h of LibriSpeech) with speech synthesized by six open-source TTS systems, about 4.8k h in total. On LibriSpeech test-clean (cross-sentence) it reaches 2.45% WER and 0.641 speaker similarity (MELLE on the same data: 3.30% / 0.606), MOS 4.21 and the best SMOS among compared models, 4.13. The streaming version reaches 3.54% WER (SMLLE 5.14%) with about 440 ms first-packet latency and RTF 0.55.

## Other Research

**TVPara: Paralinguistic Captioning of Expressive TV-Series Speech** *First author* arXiv 2026

*Motivation* Audio LLMs struggle with paralinguistics (emotion, pace, pitch) in emotional speech amid complex acoustics, because existing corpora are mostly studio-recorded or offer only categorical labels.

*Approach* Built TVPara from TV series via LALM pre-annotation and human correction: single-utterance (about 492 h) and a new multi-utterance task (about 213 h) on how attributes change across utterances. Fine-tuning open-source LALMs on it gives a 31.26% average relative improvement under complex acoustic conditions, and the gains carry over to an unseen series and to cleaner cross-domain speech.

**Bayesian WeakS-to-Strong from Text Classification to Generation** *Co-first author* **ICLR 2025**

*Motivation* If AI eventually surpasses humans, how can humans supervise it? We proposed drawing on collective human wisdom, using several supervisors to simulate the diversity of human opinions. With no such AI available then, small models stood in for humans and a larger model for superintelligence. That expectation has since come true: AI now surpasses humans in mathematics, coding and other fields.

*Approach* An ensemble of weak models acts as the supervisors, and Bayesian evidential learning (a Dirichlet prior) estimates the confidence of their labels to guide the strong model. Token-level soft labels extend the method to generation, combined with (conservative) DPO on weak-model preferences. Performance gap recovered reaches 0.828 on SciQ classification (baseline 0.754) and 0.668 on SLURP generation, 0.705 with conservative DPO.

*My role* Proposed the core idea; implemented the method and ran the classification experiments that gave the first proof of feasibility. The extension to generation was done by collaborators.

**SWIM: EEG-Based Auditory Spatial Attention Decoding with a Short-Window CNN and Mamba** *First author* **IEEE SLT 2024**

*Motivation* Auditory spatial attention decoding (ASAD) tells from EEG whether a listener attends to the left or right speaker without clean speech, a key step for neuro-steered hearing aids. It must react to attention switches in short windows while using long-term EEG.

*Approach* A short-window CNN (SW_CNN) with EEG data augmentation, multitask training and model combination reaches 84.9% accuracy in the leave-one-speaker-out setup on KUL, 4.9 points above the previous best. SWIM adds Mamba to integrate long-term history and reaches 86.2%, a 31.0% relative error reduction; new best results in all three setups.

## Publications

* *equal contribution*


[1] **Ziyang Zhang***, S. Li*, Y. Wang, Z. Cui, B. Zhou, W. Wu, C. Zhang. MMTA: Benchmarking Multimodal Temporal Analysis with Time Series, Text, and Vision. **NeurIPS 2026**. arXiv

[2] W. Wu, **Ziyang Zhang**, L. Liu, X. Xu, et al., C. Zhang. SciTS: Scientific Time Series Understanding and Generation with LLMs. **ICLR 2026**. paper

[3] Z. Cui*, **Ziyang Zhang***, G. Sun, W. Wu, C. Zhang. Bayesian WeakS-to-Strong from Text Classification to Generation. **ICLR 2025**. paper

[4] **Ziyang Zhang**, A. Thwaites, A. Woolgar, B. Moore, C. Zhang. SWIM: Short-Window CNN Integrated with Mamba for EEG-Based Auditory Spatial Attention Decoding. **IEEE SLT 2024**.

[5] **Ziyang Zhang**, S. Li, W. Wu, C. Zhang. TimeOmni-o1: Bidirectionally Coupling Time Series Foundation Models and LLMs for Multimodal Reasoning and Forecasting. OpenReview

[6] S. Li*, **Ziyang Zhang***, W. Wu, M. Liu, C. Zhang. Time-JEPA: Grounding Video World Models with Physical Time Series. OpenReview

[7] **Ziyang Zhang**, W. Yu, W. Wu, X. Xu, C. Zhang. TVPara: Paralinguistic Captioning of Expressive TV-Series Speech in Complex Acoustic Environments. arXiv 2026. arXiv

[8] **Ziyang Zhang**, Y. Gao, X. Xu, B. Li, W. Wu, C. Zhang. Bayesian Speech Synthesizers Can Learn from Multiple Teachers. arXiv 2025. arXiv

[9] Intern-S2 Team (incl. **Ziyang Zhang**). Intern-S2-Preview: Scientific Agentic Foundation Model. Technical report, 2026. arXiv

[10] Intern-S1 Team (incl. **Ziyang Zhang**). Intern-S1: A Scientific Multimodal Foundation Model. Technical report, 2025. arXiv